

Spatio Temporal Traffic Violation Detection using YOLOv8 and GNSS Mapping System

**Real-Time Traffic Monitoring with Edge Devices
and AI Vision**

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Introduction

The road transport scenario in India is marred by the vast ignoring of safety standards and it results in thousands of deaths annually. Although big cities have begun application of smart surveillance infrastructure, Tier-2 cities, IT parks, and campuses still have safety concerns as cloud-based enforcement systems are expensive and complex to install.

The centralized solutions that rely on the internet are inappropriate in the infrastructure-limited, rural, or semi-urban areas. Manual enforcement in these regions is ineffective, unscalable, and not very efficient.

To this end, we introduce SPATIO-TEMPORAL TRAFFIC ENFORCER: an offline-compatible edge AI system that can detect important traffic offenses in real-time with hybrid deep learning models and can run completely on a Raspberry Pi at a low-cost. The system provides alerts through GSM or Twilio, as well as GPS-tagged evidence without using cloud support.

Problem Statement

Riding without helmets, triple riding, or using cell phones behind the wheels and riding in the wrong direction remain some of the greatest safety hazards on the Indian roads. Such crimes are highly skewed towards Tier-2 cities, educational institutions and campuses, and IT parks and complexes where the law is not enforced adequately or at all.

Of major concern is:

- Lack of specialized surveillance infrastructure beyond metropolitan places
- Manual monitoring is time consuming and man power intensive
- Lack of awareness and noncompliance with the traffic safety regulations in the areas of private or semi-public use
- Lack of real time monitoring systems of common traffic offenses
- Smart city-level systems are not affordable and complicated enough to be implemented in smaller institutions or towns
- Incident log or alert systems were not available in areas that did not have internet access or central databases

This has led to the fact that minor and yet threatening violations are rarely punished, and this means that the risk of accidents, injuries, and continued ignorance of the safety of the road in such areas rises.

Purpose

The core aim of the project is to identify and notify the authorities on regular traffic offenses in real time particularly in places with poor infrastructural set up or where the internet is not easily accessible. The objective of the system is to:

Identify numerous infractions of live video feed:

- Helmetless riding
- Triple riding
- Usage of mobile phones when driving
- Wrong-way driving
- Retrieve metadata such as GPS coordinates and timestamp of every violation

- Make the best interactive dashboard
- Runs completely offline or online on Raspberry Pi system
- The most important innovation is a single task of detection of objects, determination of their pose, and tracking of the objects in a minimalist, low power, and offline-capable enforcement system that can be deployed in the real world.

System Architecture

Hardware Configuration

- Raspberry Pi 4 Edge AI processor for detection & alerting
- IP Camera (RTSP) Provides live video input
- Neo-6M GPS Module Provides GPS coordinates for each event
- Power Source Ensures 24/7 power via power bank or UPS

Software Pipeline

- YOLOv8n: Detects helmets, people, bikes, and number plates
- ByteTrack: Maintains consistent IDs for tracked objects
- MediaPipe Pose: Extracts body keypoints for mobile phone detection
- OpenCV + Python: Handles video processing and integration
- JSON or CSV Logger: Stores structured logs for all events

Violation Detection Modules

1.Detection of Helmet

Two-wheelers have riders detected by a model trained on the YOLOv8n model specifically on helmet detection. ByteTrack aids in delivering similar tracking of objects in the frames. In case an individual is found on a bicycle without a helmet, a violation is registered.

- Reason: Bike and rider are present, and helmet not recognized
- Detection Accuracy: 87mAP
- Frames per second: 12 FPS on Raspberry Pi

2.Triple Riding Detection

The number of people on one bike is flagged as the triple riding by using YOLOv8 and ByteTrack. When additional than two individuals are monitored in the overlapping of the same bike, it is considered as a violation.

- Logic: Individuals on same bike > 2
- Tracking: ByteTrack does not flicker and duplicate itself
- Output: Snapshots of violations and time-stamped metadata

3.Mobile Phone Use Detection

Using a phone while driving is detected by analyzing body pose instead of just object appearance. MediaPipe Pose estimates keypoints (hands, ears, elbows), and if a hand remains near the ear for several frames, the system infers phone usage.

- Detection Mechanism: Pose-based, not object-based
- Filtering: Temporal smoothing avoids false positives
- Daytime Accuracy: ~88%

4 .Wrong-Way Driving

Vehicles moving against the correct direction of traffic are detected using object tracking data and motion vectors. The camera's field of view defines acceptable directions.

- Logic: Entry and exit paths mapped per camera angle
- Tracking: ByteTrack ensures accurate path detection
- Use Case: One-way roads, IT parks, campus exits

5. Alert and Logging System

- GPS Logging
- Each violation is associated with its geolocation and time of occurrence using the Neo-6M GPS module. Data is saved locally in JSON format.
- Fields: Violation type, plate, latitude, longitude, time, image

Example:

```
{  
  "violation": "helmetless",  
  "plate":  
    "MH12BC7789",  "lat":  
    "18.5404",  
  "lon": "73.8532",  
  "time": "2025-07-20 12:04:53",  
  "image": "helmet_15.jpg"  
}
```

6. Interactive Dashboard

A locally hosted dashboard helps visualize all violation data and supports traffic enforcement personnel.

Key Features:

- Violation Log History:
Tabular data: time, location, type, snapshot
- Map View:
Embedded map with location pins for each violation
Clickable markers show evidence
- Violation Counter:
Daily, weekly, monthly
counts Filter by violation type
- Graphical Analytics:
Bar chart showing frequency
- Snapshot Viewer:
Image/video browser for all recorded events
- E-Challan Preview:
Generates a challan-style view with violation and fine info

Model Training and Evaluation

Helmet Detection	5000 images	87% accuracy
Number Plates	2,500 pictures	81% accuracy

Google Colab and Roboflow were used to train models to use real-world labeled datasets. To enhance frame-to-frame tracking, ByteTrack was incorporated, and MediaPipe was calibrated on various postures and diverse riding conditions.

Testing and Results Field

The system has been put into real test in:

- College and School zones
- Roads and gates in IT park
- Intersections and one-way routes in the tier-2 cities

Performance Metrics:

- Real time alerts at ~11 FPS
- Full work offline, does not require the internet
- GPS-marked books and plate recognition
- Record creation in the form of snapshots and SMS based records

Creativity and Originality

The project is among the earliest traffic violation detection systems based on YOLOv8, ByteTrack, and MediaPipe Pose which is offline and online capable and does not require cloud-servers .

What is special about it:

- Offline deployment of edge-AI Completely offline
- Hybrid visual and pose-based detection use
- Activities in low resource settings
- Made to suit schools and Tier-2 areas
- Compact and low-power consuming

Conclusion

This can act as the cheap and effective replacement of smart cities surveillance in traffic policing. Targeted at under-monitored areas, including school campuses, IT parks, and Tier-2 cities, it monitors important violations, provides evidence in GPS-based format, and alerts the authorities in the real time. Its good accuracy of detection, portability, and remote capabilities make it a viable safety device with a robust potential of being used in the mid-urban and semi-urban settings in India.